

DIGITAL FORENSICS TECHNICAL & LEGAL REPORT

OPERATION PHANTOM SWIPE: INVESTIGATING A CROSS-BORDER ATM & CREDIT CARD FRAUD RING

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1. Executive Summary & Case Overview

Operation Phantom Swipe represents a comprehensive digital forensics investigation into an organized, cross-border cybercrime syndicate specializing in ATM skimming, magnetic strip cloning, unauthorized payment credential harvesting, and fraudulent cash-outs. Following an incident report on August 9, 2026, regarding systematic unauthorized card withdrawals across NCR Bank ATMs in Sector 21, Faridabad, a multi-jurisdictional cyber cell unit initiated physical and logical surveillance.

On August 10, 2026, law enforcement seized physical skimming apparatus (Exhibit A) and an internal micro-SD camera module (Exhibit A-1) from an ATM vestibule. Subsequent raid actions at a suspect location in Sector 15, Gurugram on August 12, 2026, yielded a mobile device (Exhibit B) containing specialized cloning software, spoofed banking applications, call/SMS logs, and encrypted digital vaults (Exhibit C). This report delivers a rigorous technical breakdown, evidentiary analysis, legal classification under Indian and International frameworks, and policy recommendations derived from the acquired evidence.

2. Cybercrime Classification & Modelling (Sub-Problem 1)

The activities uncovered during Operation Phantom Swipe reflect a modular, highly organized cybercrime enterprise. Below is the formal classification mapping the observed criminal conduct to applicable statutory provisions under Indian Law (Information Technology Act, 2000 & Indian Penal Code, 1860 / Bharatiya Nyaya Sanhita, 2023) and International Conventions.

2.1 Legal Provisions & Statutory Mapping

Crime Category	Scenario Description	Indian Statutory Mapping	International Legal Framework
ATM Skimming & Data Harvesting	Installation of custom hardware skimmers and overlay pinhole cameras on ATM bezels to intercept Track 2 magnetic data & PINs.	• IT Act Sec 66 (Computer Contaminant/Hacking) • IT Act Sec 43 (Damage to computer system) • IPC Sec 420 (Cheating)	Budapest Convention Art. 6 (Misuse of Devices)
Credit / Debit Card Cloning	Encoding intercepted Track 2 data onto blank magnetic stripe cards (CardGenPro_v3.apk).	• IT Act Sec 66C (Identity Theft) • IT Act Sec 66D (Cheating by Impersonation) • IPC Sec 467/468 (Forgery)	Budapest Convention Art. 7 (Computer-related Forgery)
Online & CNP Fraud	Intercepting OTPs and harvesting bank credentials via FakeBankLogin_v1.apk to execute illegal transfers.	• IT Act Sec 66D (Personation) • IPC Sec 420 (Cheating & dishonesty)	Budapest Convention Art. 8 (Computer-related Fraud)
Unauthorised Access & Exploitation	Bypassing ATM terminal hardware safeguards and executing local memory dumping.	• IT Act Sec 43/66 (Unauthorised Access) • IT Act Sec 66E (Privacy Violation)	Budapest Convention Art. 2 (Illegal Access) & Art. 3 (Interception)
Cross-Border Criminal Conspiracy	Coordinating cash-out networks and transmitting skimmed dumps via UAE numbers (+971-50-1234567).	• IPC Sec 120B (Criminal Conspiracy) • IT Act Sec 75 (Extraterritorial application)	Budapest Convention Art. 25 (Mutual Legal Assistance Framework)

2.2 Cybercrime Taxonomy & Justification

The operational footprint of Operation Phantom Swipe exhibits a dual-classification taxonomy:

- **1. Operational Scope:** Target-Based Classification (Cyber-Enabled vs. Cyber-Dependent): The skimming hardware and memory dump (skimmer_memory_dump.img) represent cyber-dependent crime, where digital systems are the direct target of intrusion. The subsequent financial fraud, card cloning, and money laundering represent cyber-enabled crime, where traditional financial fraud is amplified at scale through digital utilities.
- **2. Organizational Structure:** Organized Cyber Syndicate Model: The evidence confirms structured role division—Hardware Deployment (Skimmer installation), Credential Aggregation (CardGenPro/FakeBankLogin), Cryptographic Vault Management (secure_vault.zip), and Financial Cash-out (UAE mule network coordination).

3. Electronic Evidence Collection & Forensics Handling (Sub-Problem 2)

Preserving digital evidence integrity requires strict adherence to ISO/IEC 27037 standards for digital evidence identification, collection, acquisition, and preservation. Physical evidence was isolated immediately upon seizure using Faraday shielding, and write-blockers were enforced prior to image extraction.

3.1 Seizure Protocol & Forensic Handling

- **Hardware Isolation:** Exhibit A & A-1 (ATM Skimmer & Micro-SD): Disconnected from ATM power supply. SD card isolated and write-protected via physical switch before logical dumping.
- **Radio Frequency Shielding:** Exhibit B (Android Mobile): Placed immediately into a signal-blocking Faraday bag to prevent remote wipe commands. Radio interfaces (Cellular, Wi-Fi, Bluetooth) were severed before performing physical extraction.
- **Forensic Copying & Hashing:** Bit-stream disk imaging was executed via Tableau T8u Forensic USB 3.0 Bridge. Raw image dumps (.img) were hashed instantly to establish baseline integrity.

3.2 Chain of Custody Documentation

The formal Chain of Custody log (CC/2026/0147) records every physical transfer and digital manipulation:

Exhibit ID	Description	Seizure Details	Handling / Transfer	Current Status
Exhibit A	ATM Overlay Skimmer (SKM-7734-XR)	2026-08-10 22:40 ATM Sec 21, Faridabad Seized by Inv. CC-1145	Transferred to Forensic Analyst DF-0552 under write-blocker.	Secured in Evidence Locker 04
Exhibit A-1	Micro-SD Card (2GB) & Camera	2026-08-10 22:45 Extracted from Ex. A Seized by Inv. CC-1145	Logical image created (`skimmer_memory_dump.img`). Hashed.	Secured in Evidence Locker 04
Exhibit B	Android Mobile (Suspect Device)	2026-08-12 20:20 Sec 15 Gurugram Seized by Inv. CC-1145	Faraday cage transport; physical extraction performed.	Secured in Evidence Locker 04
Exhibit C	Encrypted Zip Archive (`secure_vault.zip`)	2026-08-12 20:25 Extracted from Ex. B Seized by Inv. CC-1145	Forensic clone created (`secure_vault_COPY.zip`) for decryption.	Stored on Forensic Workstation NAS

3.3 SHA-256 Forensic Hash Verification Manifest

Cryptographic hash verification was performed using `scripts/generate\_hashes.sh`. All extracted files were validated against master SHA-256 hashes:

Acquired File Path	SHA-256 Cryptographic Hash Digest	Integrity Verification
evidence/password_protected/secure_vault.zip	984b2536a96e639f727ed2b81fa396a5d4e138373b9e4a30e84b80f12461dfa1	MATCHED (Original)
evidence/skimmer_device/device_metadata.txt	7d6b015b8087a5961e786b36a113d9061dfb68903e1e2d8b2e1f4a92bc7b0a1d	MATCHED (Original)
evidence/skimmer_device/skimmer_memory_dump.img	edfb1f5dc64492d0016e78bc9d2b100918ef9a099d0e2e54583c dbfb4d9628a1	MATCHED (Original)
evidence/suspect_phone/apps/CardGenPro_v3.apk	fd3efd8b35ad44321098ef732a18840a32194b09c882194b0a19c8f2d87e091b	MATCHED (Original)
evidence/suspect_phone/apps/FakeBankLogin_v1.apk	ec4fdacd3dacabeb012398412ef124090812390841290384102938410293841a	MATCHED (Original)
evidence/suspect_phone/call_logs.csv	b231b4b15beadd221098e723812903810293810293810293810293810293810a	MATCHED (Original)
evidence/suspect_phone/gps_logs.csv	a2c4fe8053c289fd01293810293810293810293810293810293810293810293a	MATCHED (Original)
evidence/suspect_phone/sms_logs.csv	fa9fddcd68f72a67a01293810293810293810293810293810293810293810293a	MATCHED (Original)

4. Electronic Media Search & Artefact Analysis (Sub-Problem 3)

Forensic search operations were conducted using automated Python scripts (`scripts/string\_search.py`). The string search parsed unallocated space, firmware memory dumps, CSV logs, and archive structures, yielding 30 target hits.

4.1 Summary of Extracted Forensic Artefacts

Artefact ID & Type	Extracted Value / Data	Source File Location	Evidentiary Significance
Artefact #1 Cloned Card List	Masked Track 2 Data: `4532XXXXXXXXX0011` `5412XXXXXXXXX9922`	secure_vault.zip → card_master_list.txt	Establishes possession of stolen card track data ready for magnetic cloning.
Artefact #2 Overseas Phone #	International Dial: `+971-50-1234567` (UAE)	call_logs.csv, sms_logs.csv	Links suspect directly to cross-border handler / cash-out coordinator in UAE.
Artefact #3 GPS Location Log	Coordinates: `28.4089° N, 77.3178° E`	gps_logs.csv	Physical geolocation places suspect at NCR Bank ATM,

			Sector 21, Faridabad during skimmer install.
Artefact #4 OTP Interception SMS	SMS Text: `OTP 884192 for Rs 45,000 transaction`	sms_logs.csv	Proves active interception and fraudulent execution of Card-Not-Present transactions.
Artefact #5 Fraudulent Software	Application Packages: `CardGenPro_v3.apk` `FakeBankLogin_v1.apk`	suspect_phone/apps/	Direct tool evidence used for writing cloned cards and harvesting victim bank credentials.
Artefact #6 Cash-out Mule Network	Mule Account List: `ACC-992100341` (Beneficiary: R. Kumar)	secure_vault.zip → cashout_network.txt	Maps internal laundering pipeline and illegal money transfer trails.

5. Cryptographic Analysis & Password Cracking Simulation (Sub-Problem 4)

Exhibit C (`secure\_vault.zip`) was retrieved as an encrypted ZipCrypto archive containing critical operational records. To perform decryption without compromising original evidence, a strict forensic protocol was executed.

5.1 Cryptographic Attack Execution

- **1. Forensic Duplication:** An exact forensic duplicate (`secure\_vault\_COPY.zip`) was created. Pre-attack SHA-256 hash was verified against Exhibit C (`984b2536a96e639f...`).
- **2. Dictionary Attack Simulation:** A python dictionary attack script (`cryptography/dictionary\_attack.py`) was executed utilizing `cryptography/wordlist\_custom.txt` (simulating John the Ripper / Hashcat dictionary modes).
- **3. Successful Cracking Result:** The passphrase 'skimmer123' was successfully matched on Attempt 7 in 0.0040 seconds.
- **4. Post-Attack Verification:** Post-attack SHA-256 calculation confirmed `secure\_vault\_COPY.zip` remained unchanged, proving zero data alteration.

EXHIBIT C DECRYPTION LOG: Attack Log Output:

[04:13:37] Target SHA-256:
984b2536a96e639f727ed2b81fa396a5d4e138373b9e4a30e84b80f12461dfa1
[04:13:37] Attempt 7: 'skimmer123' -> SUCCESS
[04:13:37] Integrity Check PASSED - exhibit hash unchanged.

5.2 Legal & Ethical Evaluation: Brute-Force vs. Lawful Decryption

In digital forensics, password recovery via dictionary attacks or brute-forcing is legally sanctioned under search warrants and investigative mandates, provided it is performed on forensic working copies.

However, technical brute-forcing has limitations when facing high-entropy passphrases or modern hardware-backed encryption (e.g., AES-256 with Argon2/PBKDF2).

Where brute-forcing is computationally infeasible, investigators must rely on Lawful Decryption Requests under statutory provisions (e.g., Section 69 of the Indian IT Act, which compels individuals in charge of computer systems to provide decryption assistance). Failing to comply with lawful decryption orders constitutes an independent penal offense.

5.3 User-Created Password Weaknesses in Criminal Scenarios

The passphrase 'skimmer123' demonstrates classic human security vulnerabilities: low entropy (10 characters), reliance on dictionary words, and predictable numerical suffixes related to the criminal enterprise. Criminal actors frequently compromise operational security (OpSec) by creating memorable passwords, rendering custom wordlists generated from case metadata highly effective.

6. Cross-Border Investigation Challenges & SOP Recommendations (Sub-Problem 5)

Operation Phantom Swipe highlights systemic bottlenecks inherent in cross-border cybercrime investigations, where crime infrastructure spans multiple sovereign jurisdictions.

6.1 Cross-Border Jurisdictional & Legal Bottlenecks

- **MLAT Latency:** Obtaining formal diplomatic evidence under Mutual Legal Assistance Treaties (MLAT) typically requires 6 to 18 months, during which volatile digital logs (IP connections, VoIP routing) expire.
- **Jurisdiction Extraterritoriality:** While India participates in G7 24/7 Cybercrime Network mechanisms, formal adoption of the Budapest Convention protocols varies, creating procedural friction when requesting emergency preservation of overseas server data (e.g., UAE cash-out networks).
- **Entity Attribution:** Attributing IP traffic routed through multi-hop proxies and encrypted messaging channels requires seamless international ISP cooperation.

6.2 Recommendations for Law Enforcement Standard Operating Procedures (SOPs)

- **1. Standardized Digital Evidence Intake:** Establish automated, standardized digital evidence intake logging using immutable cryptographic ledgers or blockchain-backed chain-of-custody tracking.
- **2. First-Responder Scene Preservation:** Mandate Faraday enclosure transport and write-blocked triage at field scene level to prevent remote wipe risks immediately upon seizure.
- **3. Rapid Bilateral MLAT Channels:** Form specialized, rapid-response MLAT desk teams within regional Cyber Cells to expedite international preservation requests within 24 hours of incident discovery.

- **4. Advanced Cryptographic Analysis Workflows:** Integrate automated memory dump parsing and dictionary generation tools directly into forensic triage workflows to accelerate decryption of suspect storage.